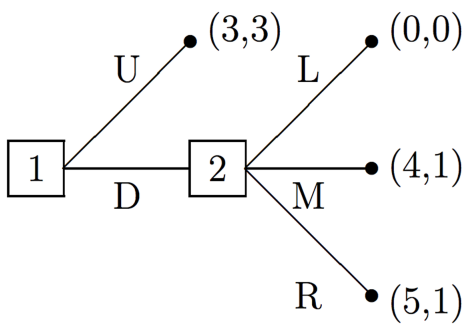
6CCSACCM Cooperation, Coordination and Multi-Agent Systems:
Repeated and extensive-form games

1. Consider the extensive form game with players 1 and 2 shown in the following figure:



   (a) Write down the normal-form representation (i.e., the payoff matrix) of the game. Find all pure strategy Nash equilibria.

   Solution. The payoff matrix of the game is given by:

Players 1,2	L	M	R
U	3,3	3,3	3,3
D	0,0	4,1	5,1

   There are three pure strategy Nash equilibria in this game: (U, L), (D, M) and (D, R).

   (b) One of the pure strategy Nash equilibria is NOT a subgame-perfect Nash equilibrium. Which one? Explain your answer.

   Solution. (U, L) is not a subgame perfect equilibrium, as playing L is not a best response for player 2 in the subgame rooted at node 2.

2. Consider the following game with partial information.

   Two players start by putting £1 in the pot.

   One card is dealt to player P_1. There is 0.5 chance of a high card, H, and 0.5 chance of a low card, L.

   Player P_1 knows which card s/he has been dealt, and can either see or raise.

   If player P_1 sees, then P_1 gets the pot if the card is H and P_2 gets the pot of the card is L.

   If P_1 raises, then s/he must put in £k more money in the pot.

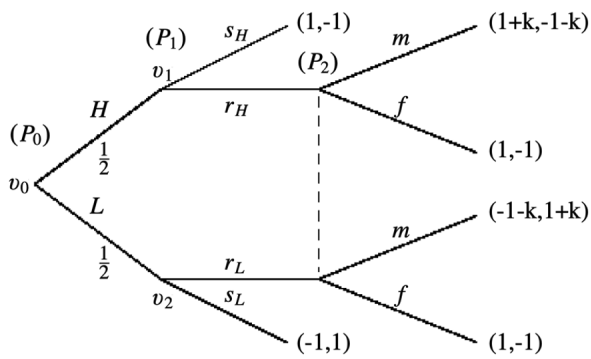
Then player P_2 must decide whether to fold or meet (call) the raise.

If P_2 folds, then P_1 gets the pot.

If s/he meets, then s/he must put £k more money in the pot. Then P_1 gets the pot if the card is H and P_2 gets the pot of the card is L.

Describe the extensive-form version of the game using a tree representation. Indicate information sets.

Solution.



One card is dealt to player P1. There is 0.5 chance of a high card, H, and 0.5 chance of a low card, L. We represent this by saying that the root node v_0 is owned by "nature" or "chance", P_0.

Player P_1 can either see or raise. Since P_1 knows which card s/he has been dealt, the choices are labelled s_H and r_H at v_1 (corresponding to the card H) and s_L or r_L at v_2 (corresponding to the card L).

If player P_1 sees, then P_1 gets the pot if the card is H and P_2 gets the pot of the card is L. Thus, the payoffs are $u(s_H) = (1, -1)$ and $u(s_L) = (-1, 1)$.

If P_1 raises, then s/he must put in £k more money in the pot, and player P_2 must decide whether to fold or meet (call) the raise.

If P_2 folds, then $u(*, r_*, f) = (1, -1)$.

If s/he meets, then s/he must put £k more money in the pot. Then, the payoffs are $u(H, r_H, m) = (1 + k, -1 - k)$ and $u(L, r_L, m) = (-1 - k, 1 + k)$.

The information set $I = \{v_3, v_4\}$ is the set of vertices where P_1 has raised and P_2 does not know whether the card is H or L. Player P_2 must make the same choice at both vertices in I.

3. Consider a repeated game such that with probability p the game continues to the next round and with probability $(1 - p)$ it ends.

	Cooperate	Defect
Cooperate	2,2	0,5
Defect	5,0	1,1

(a) The game starts in round 1. In odd rounds both players cooperate and in even rounds both defect.

What is the expected total future payoff (starting at the beginning of the game) for each player, when the game is forecast to be played as described above?

Solution. In odd rounds each will get a payoff 3 while in even rounds each gets a payoff 1. Thus, the expected total future payo for each player is

$$2 + p + 2p^2 + p^3 + 2p^4 + p^5 + \dots$$

(b) Let p^* be the threshold such that when $p \geq p^*$, cooperation is sustainable as a subgame-perfect equilibrium by the trigger strategy (under which each player cooperates as long as the other does and defects forever after if either player deviates), and when $p < p^*$ it is not sustainable. Find p^*.

Solution. The continuation payoff from indefinite cooperation is

$$2 + 2p + 2p^2 + \dots = 2 + \frac{2p}{(1-p)}$$

The continuation payoff from defecting (and thus continued defection) is

$$5 + p + p^2 + \dots = 5 + \frac{p}{(1-p)}$$

In order to sustain cooperation, we need

$$2 + \frac{2p}{(1-p)} \geq 5 + \frac{p}{(1-p)} \Rightarrow p \geq \frac{3}{4}$$

Hence, $p^* = \frac{3}{4}$.